

Doc. 311 – Four onto Three

How does a 4D lattice arrive at three spatial dimensions?

Johann Pascher

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Contents

Doc. 311 – Four onto Three	2
A The three routes	3
A.1 Route 1 — Compactify	3
A.2 Route 2 — Project	3
A.3 Route 3 — The fourth is not space	3
B Routes 1 and 3: each incomplete on its own	4
B.1 What closes the gap	4
B.2 The sharper question	4
C Route 2 in detail: what projecting presupposes and costs	6
C.1 Projection needs a standpoint outside	6
C.2 What happens to the 24 vectors	6
C.3 Where the irrational numbers come from	7
C.4 The consequence for other constructions	7
D The reversal: where φ is found, the direction is no longer free	8
D.1 Where φ stands in the corpus	8
D.2 φ is not a choice here but an eigenvalue	8
D.3 And the window as well	9
D.4 The juxtaposition	9
D.5 What this means	9
E Two conditions instead of three routes	10
E.1 What becomes transferable	10
E.2 What does not become transferable	10
E.3 Where the two nevertheless meet	10
E.4 The rule	11
E.5 What this means for the examples	11
F The finding that decides the matter	12
F.1 Why the pyramid is fundamental in three dimensions	12
G What follows for the invariants	14
G.1 Which packing density belongs in K^{-36} ?	14
H Application: constructions with four spatial lattice directions	16
I The two conditions, applied to string theory	17
I.1 Condition 2 is met exemplarily	17
I.2 Condition 1 is met by half	17
I.3 What the landscape thereby is	18
I.4 The asymmetry that connects the two	18

I.5	What this does not say	18
J	Verification script	19
K	References	20

Doc. 311 – Four onto Three

What this is about

A lattice in $\mathbb{R}^4$ has four directions on an equal footing. Measurement takes place in three. How does one get from the four to the three?

There are exactly three routes, and their costs differ greatly. This note walks through them, condenses them at the end into two conditions, and applies those conditions to three cases: the author's own construction, generic projection constructions, and string theory.

Status: working note. Not part of the A-Series. The chapters on the routes and on the decomposition finding are computed (`ffgft_311_vier_auf_drei.py`); the chapter "Application" poses a question and makes no assertion.

Chapter A

The three routes

A.1 Route 1 — Compactify

The fourth direction becomes a circle of radius R . Kaluza 1921, Klein 1926.

What it buys. On a circle only integer windings are admitted, and from that a mode tower arises with $m_n \propto n/R$. The mode index *is* a scale. Nothing has to be attached.

What it costs. R is a free modulus. Whoever does not say what fixes R has gained a scale and acquired a parameter — that is, nothing. This is the price string theory pays with its landscape.

The modulus disappears only when R is bound to another quantity. In FFGFT this happens through $\tilde{T} \cdot m = 1$: once the mass is fixed, the cycle is fixed, and nothing is left to tune.

A.2 Route 2 — Project

The 4D lattice is projected onto a 3D section. The standard route for quasicrystals, cut-and-project.

What it costs. The projection has a window parameter, and the lattice invariants do not survive it. For D4 onto the first three coordinates:

$$24 \text{ shortest vectors} \longrightarrow 18 \text{ distinct images.}$$

Six pairs coincide. What arrives after projection is no longer a lattice with kissing number 24, but an aperiodic point pattern with different invariants.

A.3 Route 3 — The fourth is not space

The fourth coordinate carries a different quantity: time, mass, charge, an internal index. Then only three directions are space, and there is nothing to reduce.

What it costs. One must say *what* it carries, and the statement must bear weight. This is no formality: a fourth direction called “mass” without anything binding it to a mass is merely a label.

Chapter B

Routes 1 and 3: each incomplete on its own

The two look alike because both take the fourth direction out of space. But they do different things, and the difference is precisely the point at which a construction holds or does not.

Route 1 makes the direction compact and leaves open what it carries. Pure Kaluza-Klein. A circle of radius R enforces integer windings, from that a mode tower $m_n \propto n/R$ arises, and the mode index is a scale. But R itself is free. One has gained a scale and bought a parameter along with it.

Route 3 says what the direction carries and leaves open whether it is compact. It is called "mass" or "charge" or "internal index". Without compactness this is a label: nothing enforces integer values, without integrality there is no mode tower, and without a mode tower no scale.

Each is incomplete on its own, in opposite ways:

Route 1: mechanism without meaning

Route 3: meaning without mechanism

B.1 What closes the gap

Both at once, and the binding between them.

	compact	bound	result
Kaluza-Klein	yes	no	scale present, but free modulus
mere label	no	in name only	no mode tower, no scale
FFGFT	yes	yes, via $\tilde{T} \cdot m = 1$	scale without modulus

The reciprocity closes both gaps in one step. The mode tower acquires a meaning because the direction is the mass direction. And the free radius disappears because R is no longer choosable once m is fixed — the cycle is the mass, not a container for it.

B.2 The sharper question

The real distinction is therefore not "which of the three routes", but:

Is the fourth direction compact **and** bound to something — or only one of the two?

Whoever only compactifies has a modulus to stabilise. Whoever only names has a word. Both together and bound to each other is the only case in which a scale arises without a parameter following behind.

This is also the touchstone for other people's constructions, and it can be put in a single question: *what fixes the radius?* If no answer comes, the scale has merely been displaced.

Chapter C

Route 2 in detail: what projecting presupposes and costs

Route 2 differs from the other two not in degree but in kind.

C.1 Projection needs a standpoint outside

Compactifying is a property of the object: the circle is there or it is not, regardless of who is looking. Interpreting the fourth direction as the mass direction is a statement about the structure.

Projecting is a map **from outside**. It needs a cut direction and a window, and both are chosen by the observer. They say nothing about the lattice; they say something about the access to it.

This is a different type of free parameter from the radius in Route 1. A radius is a feature of the object and can be bound to another quantity of the object — exactly what $\tilde{T} \cdot m = 1$ achieves. A cut direction cannot be bound. It can only be fixed, and the fixing remains a stipulation.

C.2 What happens to the 24 vectors

Under projection onto the first three coordinates:

24 vectors	→	18 images
12 FCC vectors ($x_4 = 0$)	→	12 images, length $\sqrt{2}$, untouched
12 winding vectors	→	6 images, length 1, merged in pairs

For $(1, 0, 0, +1)$ and $(1, 0, 0, -1)$ have the same image $(1, 0, 0)$. The difference between two opposite windings disappears.

And the image contains two different lengths, 1 and $\sqrt{2}$. The image is no longer a lattice with a uniform kissing number; packing density in the usual sense loses its meaning there.

This is the real loss. Under compactification the twelve winding neighbours remain distinguishable — $+1$ and -1 are different windings and carry different quantum numbers. Under projection they are made indistinguishable. What leads out of space is erased rather than reinterpreted.

C.3 Where the irrational numbers come from

For a rational cut direction, say a coordinate axis, finitely many image lengths remain — here two. The shortening of the winding vectors by $1/\sqrt{2}$ is irrational, but it repeats.

In genuine cut-and-project the cut direction is chosen irrational, and then the projection lengths are generically irrational. For the direction $(1, \varphi, 0)$:

$$\begin{array}{ll} (1, 0, 0) & \longrightarrow 0.525731 \\ (0, 1, 0) & \longrightarrow 0.850651 \\ (1, 1, 0) & \longrightarrow 1.376382 \\ (1, -1, 0) & \longrightarrow -0.324920 \end{array}$$

Every length a new number, and the image is **dense**. Only the window makes it discrete again — the second free parameter.

This is exactly how quasicrystals arise. And exactly how φ enters there:

The stretching factors are irrational, yes. But they are properties of the **chosen direction**, not of the lattice. Choose differently and they are different numbers.

C.4 The consequence for other constructions

Whoever brings φ in through a cut-and-project construction has put it in. It does not come from the lattice but from the choice of projection direction. That is admissible, provided it is said — and it is something fundamentally different from a number that follows from the structure.

The touchstone from the previous chapter thus acquires a second question beside *what fixes the radius?*:

Who chose the cut direction, and from what?

If no answer comes, all numbers originating from the projection are properties of the observer.

Chapter D

The reversal: where φ is found, the direction is no longer free

If φ appears in cut-and-project because the cut direction was chosen accordingly — what holds when φ is not chosen but *found*?

D.1 Where φ stands in the corpus

A120 derives $\theta = 2/9$ from the redistribution of the lepton modes under a five-fold rotation. The three weights:

$p_0 = 2/9$	$= 0.2222222$	rational	— no φ in it
$p_2 = (5 - 3\varphi)/9$	$= 0.0162109$	contains φ	
$p_1 = (2 + 3\varphi)/9$	$= 0.7615669$	contains φ	
sum	$= 1.0000000$		

Remarkably, the share passing into the trivial mode is **rational**. φ sits solely in the split between the two non-trivial modes. All three carry the denominator $9 = 3^2$, the same Z_3 structure in which $2/9$ sits as an orbit address.

D.2 φ is not a choice here but an eigenvalue

In cut-and-project φ appears because the observer laid the direction that way. Here φ is the eigenvalue of the structure itself:

$$\varphi^2 = \varphi + 1.$$

A_5 is the icosahedral group. Its five-fold axes are the twelve vertices with coordinates $(0, \pm 1, \pm \varphi)$ and cyclic permutations. **The directions are completely fixed by φ** — no angle remains choosable, only which of the six axes, and A120 shows that this is a discrete two-fold choice ($2/9$ or $4/9$), not a free parameter.

The uniqueness has been tested there as well: 200 random rotation axes at 72° give values between 0.036 and 0.452, not a single one hitting $2/9$. For a three-fold axis the result is $2/5$, at 144° it is $4/9$.

D.3 And the window as well

An icosahedron has no shape parameter. Its circumradius follows from the edge:

$$|\text{vertex}| = \sqrt{1 + \varphi^2} = 1.9021130, \quad \frac{\text{circumradius}}{\text{edge}} = 0.9510565 = \frac{\sqrt{\varphi\sqrt{5}}}{2}.$$

All size ratios are determined by φ . What remains is one scale factor — and that is exactly the single quantity A085 carries as the anchor.

D.4 The juxtaposition

	cut direction	window shape	window size
cut-and-project, generic	two free angles	free	free
icosahedral	6 axes, discrete	the icosahedron, shape-fixed	one scale only

A continuum of freedoms on one side, a discrete selection plus one scale on the other.

D.5 What this means

Where φ is chosen, it has been put in.

Where φ is found, it fixes direction and window.

This is not wordplay but the difference between a parameter and a structure. A construction in which φ is set as the projection direction has a free angle. One in which φ occurs as the eigenvalue of a group has a discrete selection.

And it reverses the touchstone. Instead of only asking *who chose the direction*, one can ask:

Is there a structure whose eigenvalues enforce the direction?

If an answer comes, the direction is no longer a stipulation.

Chapter E

Two conditions instead of three routes

If φ enforces direction and window, Route 2 no longer costs any free parameters — as little as Route 1 with a bound radius. The division into three routes thereby becomes obsolete, but not entirely: a second condition remains, and it is independent of φ .

E.1 What becomes transferable

The balance of freedoms. If a structure enforces the assignment, it is no longer a stipulation, whether one calls it compactification or projection.

Route 1 with free radius	one parameter
Route 1 with $\tilde{T} \cdot m = 1$	none
Route 2 with free direction	several parameters
Route 2 with icosahedral axis	none

The distinction thus does not run between the routes, but between *enforced* and *stipulated*.

E.2 What does not become transferable

The conservation of information. A projection is by definition not injective. Even with an enforced direction, $(1, 0, 0, +1)$ and $(1, 0, 0, -1)$ merge into the same image $(1, 0, 0)$. Twelve winding vectors become six images, no matter who fixed the direction.

Under compactification this does not happen. The winding number remains distinguishable — $+1$ and -1 are different states and carry different quantum numbers, although space has three dimensions.

E.3 Where the two nevertheless meet

Under compactification the information does not vanish; it **migrates**. From position into an index. What was a fourth coordinate before is a winding number afterwards.

A projection with an enforced direction does the same — **provided the preimage pair carries a conserved index**. If it carries none, it is loss. If it carries one, it is a rebooking, and then the difference from compactification is merely one of phrasing.

E.4 The rule

Not three routes, but two conditions.

1. Is the assignment enforced by a structure or stipulated by the observer?

2. Is the lost positional information caught by a conserved index?

If both are met, it does not matter what the procedure is called. If the first fails, one has parameters. If the second fails, one has loss.

E.5 What this means for the examples

	condition 1	condition 2
Kaluza-Klein, free radius	no	yes
cut-and-project, generic	no	no
FFGFT, $\hat{T} \cdot m = 1$	yes	yes
icosahedral projection	yes	only with an index

The last row is the interesting one. It is the case in which φ removes the stipulation but the question of the index remains open — and that question cannot be answered by geometry, only by stating which quantity carries the index.

In FFGFT it is the winding number on the mass direction. In a pure projection construction the corresponding quantity would have to be named, otherwise the loss is real.

Chapter F

The finding that decides the matter

D4 does not need to be reduced at all. It decomposes by itself.

Of the 24 shortest vectors:

	count	meaning
with $x_4 = 0$	12	lie entirely in the first three coordinates
with $x_4 \neq 0$	12	six with $x_4 = +1$, six with $x_4 = -1$

And the twelve with $x_4 = 0$ are not just anything:

kissing number of D3 = FCC = 12.

FCC stands for *face-centered cubic* — the densest sphere packing in three dimensions, stacking sequence ABCABC. A sphere in the interior touches six neighbours in its layer, three above and three below: $6 + 3 + 3 = 12$. That this is the maximum goes back to the question of Newton and Gregory and was finally proved in 1953.

Lattice and packing are the same object in two languages: the shortest vectors of D3 are $(\pm 1, \pm 1, 0)$ and permutations, exactly twelve, and they point to the twelve contact points.

F.1 Why the pyramid is fundamental in three dimensions

The twelve neighbour vectors form, as a convex hull, a **cuboctahedron** — eight triangles, six squares, all vertices at distance $\sqrt{2}$. This is the neighbourhood figure of three-dimensional space.

And the same packing has a second shape everyone knows. Placing spheres into the hollows of a square layer — the orange stack — each sphere in the interior has

$$4 \text{ neighbours below} + 4 \text{ in the plane} + 4 \text{ above} = 12$$

Exactly the kissing number. **The square pyramid is FCC**, merely stood up differently: the view runs along the cube diagonal instead of along the edge. Hexagonal layers in the sequence ABCABC and square layers with hollows describe the same lattice.

The fundamental point is uniqueness. In three dimensions there is exactly **one** densest packing — the Kepler conjecture, shown by Hales in 1998 and formally verified in 2014. The pyramid is thus not one building form among several, but the form the density enforces. This is why copper, gold and aluminium crystallise this way, and why the fruit stand works.

For this note two things matter in that. First, the 12 is not a conventional number but provenly maximal in 3D. Second, three-dimensional space thereby has a distinguished

neighbourhood figure against which one can measure whether a construction computes in it or not: whoever works spatially with 24 does not work in the cuboctahedron.

$$24 = 12 + 12$$

Half of the D4 kissing number is three-dimensional space. The other half leads out of it — and becomes winding neighbours as soon as the fourth direction is compact, six in each direction of circulation.

This is the lattice form of $T^4 = T^3 \times S_m^1$: three spatial circles plus the fourth direction, and the split $12 + 12$ falls out instead of being put in.

Chapter G

What follows for the invariants

The invariants of D4 are **4D quantities**. They are spatial quantities only if all four directions are space.

	D4 (4D)	D3 = FCC (3D)
kissing number	24	12
packing density	$\pi^2/16 = 0.616850$	$\pi/(3\sqrt{2}) = 0.740480$

Whoever computes with 24 as a spatial kissing number thereby implicitly asserts four spatial dimensions. Whoever has three spatial dimensions plus one other direction computes spatially with 12, and the other twelve neighbours are something else — not neighbours in space, but windings.

This is no small matter. It affects every quantity formed from the neighbourhood number.

G.1 Which packing density belongs in K^{-36} ?

From this a question follows for the author's own construction, and it had to be checked: the bulk exponent of the sector ladder (A270) is bound to a packing density. If only three of the four directions are space — is it then the 4D density or the 3D density?

	inverse density	required exponent	distance to $k^*/100$	to integer
D4 (4D)	$16/\pi^2 = 1.6211389$	35.9926	$0.098 = \mathbf{0.27\%}$	0.0074
D3 = FCC (3D)	$3\sqrt{2}/\pi = 1.3504745$	22.3836	$13.707 = \mathbf{38.08\%}$	0.3836

And at integer exponent:

$$\begin{array}{llll} \text{D4:} & (75/74)^{36} = 1.621301 & \text{against } 1.621139 & +0.010\% \\ \text{D3:} & (75/74)^{22} = 1.343538 & \text{against } 1.350474 & -0.514\% \end{array}$$

(In the notation K^{-n} with $K = 74/75$; written out as $(75/74)^n$ to keep the sign of the exponent unambiguous.)

The result is unambiguous. The 3D density does not fit. It requires exponent 22.38 instead of 36.09, i.e. 38% off, and it additionally lies close to a rounding boundary — 0.38 against 0.007 for the 4D density. Even as an isolated hit it would be weak. **The 4D density belongs in K^{-36} .**

Why this is not a contradiction

The bulk is T^4 , not T^3 . The fourth direction contributes to the packing even though it is not space — because packing happens where the torus is, and the torus has four directions.

This is admissible, but it must be said. The mass direction is a genuine torus direction with integer windings, not a caption. It counts in the packing because it is geometrically present, and it does not count in the spatial kissing number because it is not space.

The two statements thus stand side by side without contradicting each other:

$$\begin{array}{ll} \text{spatial neighbourhood} & \longrightarrow 12 \text{ (FCC, the } x_4 = 0 \text{ vectors)} \\ \text{packing of the bulk} & \longrightarrow \pi^2/16 \text{ (all four directions)} \end{array}$$

And what follows for comparable constructions

The same number, two different justifications — and typically only one of them is spoken aloud.

In FFGFT, $\pi^2/16$ is the packing density of a four-dimensional torus of which three directions are space and one carries the mass. That is a statement that can be justified and is justified.

In a construction that computes on the same lattice but treats all four directions as spatial, $\pi^2/16$ is the packing density of a four-dimensional *space*. That is an assertion about the number of spatial dimensions — and it is rarely spoken aloud, because the number appears as a neutral lattice constant.

It is not neutral. Whoever uses it says something about the number of spatial dimensions, whether they want to or not.

Chapter H

Application: constructions with four spatial lattice directions

The case that occurs in practice: a construction computes on D4 with 24 offsets, has a step counter for the iteration, and nowhere declares a time direction. From this it follows necessarily: **all four coordinates are space.**

That is an assertion about the world, not merely about the model — an object in four spatial dimensions is not a particle in our space, but an entity in a space that does not exist. Even with a matching node count it would remain open what that count has to do with a mass.

And the ways out are uncomfortable:

- One direction becomes time after the fact → then **three** spatial dimensions remain, and the spatial kissing number is 12, not 24. Everything sitting on 24 would have to be recomputed.
- Time is appended as a fifth direction → then there are four spatial dimensions, one too many.
- It stays at $4 + 0$ → then one must say how a four-dimensional spatial object acquires a mass in our space.

This is the fundamental question for any such construction, and it precedes the numerical findings: as long as it is open, any correction to individual quantities addresses only symptoms.

It is mostly not asked. Where time appears neither in the code nor in the documents nor in the correspondence, the question does not yet exist for those involved — and that is understandable, for an iteration *feels* like time without being one.

Asked, not answered. Whoever builds such a construction knows its internals; this note reads from outside. There may be answers not visible from here.

Chapter I

The two conditions, applied to string theory

The rule from the chapter “Two conditions” can be applied, and the result is not sweeping. It names exactly one place.

I.1 Condition 2 is met exemplarily

The compact directions are compact. Nothing is projected, nothing merges. Winding numbers, Kaluza-Klein momentum, winding sectors, Hodge numbers, intersection numbers — all conserved indices. The positional information of the six rolled-up directions migrates completely into quantum numbers.

This is the strength of the approach and should not be passed over. In the language of the rule: the information is rebooked, not lost.

I.2 Condition 1 is met by half

The dividing line runs through the middle of the construction.

Enforced:

$D = 10$ from anomaly freedom and the critical dimension
 $N = 1$ SUSY in 4D demands $SU(3)$ holonomy, hence Calabi-Yau
from that: $10 - 4 = 6$ compact directions

The **number** of rolled-up directions is not choosable. That is a genuine enforcement, and it is stronger than anything FFGFT can adduce for its four circles — there the choice of T^4 is marked as [STIPULATION].

Stipulated:

which Calabi-Yau manifold	($\sim 10^5$ known, total number unknown)
flux quantum numbers	($\sim 10^{500}$ combinations)
Kähler and complex structure moduli	(continuum, then stabilised)
standard embedding or not	

The number of directions is enforced. The **geometry within** is not.

I.3 What the landscape thereby is

The landscape is exactly the unmet part of condition 1.

Not an error and not a scandal, but the place where the structure stops enforcing. A precise statement of location instead of an accusation.

And it explains why the problem is so stubborn: it cannot be fixed by more computation. Where a structure does not enforce, no computation can make it do so. An additional condition from outside is needed — and that is exactly what has been sought for decades.

I.4 The asymmetry that connects the two

What is compactified are **spatial** directions. Time stays outside. By A060 it follows: no dynamical scale in the state space. All scales must come from elsewhere, and in this construction they come from the moduli.

But moduli are exactly what condition 1 does not cover.

- time outside → no scale in the structure
- scales from moduli
- moduli are the unenforced part
- landscape

The landscape arises where the scales must come from. And the scales must come from there because time stands outside. It is a single choice that produces both: roll up six spatial directions and leave time open.

I.5 What this does not say

It does not say that string theory is wrong. It meets condition 2 better than anything considered here, and its dimension count is more compellingly justified than the four circles of FFGFT.

Nor does it say that the alternative would be cheaper. A rolled-up time costs causality objections, Pauli's theorem and the thermal reading — and it is nowhere accepted so far.

What it says is exclusively this: **the landscape and the necessity of attached scales are not two problems but one**. Whoever wants to solve one must start at the other.

Chapter J

Verification script

```
import itertools, math, collections

D4 = [v for v in itertools.product([-1,0,1], repeat=4)
      if sum(x*x for x in v) == 2]
D3 = [v for v in itertools.product([-1,0,1], repeat=3)
      if sum(x*x for x in v) == 2]

print(len(D4))                # 24
print(sum(1 for v in D4 if v[3] == 0)) # 12
print(sum(1 for v in D4 if v[3] != 0)) # 12
print(len(D3))                # 12 (FCC)
print(len(set(v[:3] for v in D4))) # 18 (projection)
print(collections.Counter(
    v[3] for v in D4 if v[3] != 0)) # {1: 6, -1: 6}

print(math.pi**2 / 16)        # 0.616850 D4
print(math.pi / (3 * 2**0.5)) # 0.740480 D3
```

Extended version with all numbers of this document: `python/Dok311_Skripte/ffgft_311_vier_auf_drei.py`.

Chapter K

References

Internal documents: A060 (time as scale), A085 (anchor), A120 ($\theta = 2/9$, five-fold rotation), A270 (bulk exponent 36, P35), Doc. 275 (scale recursion depth k^*), Doc. 310 (resonance geometry as reference, concert-pitch anchor), Register 190 (R67: uncertainty of the factor 100 in the denominator of $k^*/100$).

On status: working note for the author's own use. The chapters on the routes, on the decomposition finding $24 = 12 + 12$ and on the packing density are computed; the chapter "Application" is a question and not an assertion.